12 A neutron moving with an initial velocity u has a head-on elastic collision with a stationary proton. After the collision, the velocity of the neutron is v and that of the proton is w. Taking the masses of the neutron and proton to be equal, which one of the following statements is *wrong*?

A Conservation of momentum shows that $u = v + w$.

B Conservation of energy shows that $u^2 = v^2 + w^2$.

C The fact that the collision is elastic implies that the neutron and proton move off in opposite directions with equal speeds.

D The momentum and energy equations taken together imply that the speed of the proton after the collision is the same as that of the neutron before the collision.